EXP 7: FLOW CONTROL AND DATA LINK LAYER

AIM:

Write a program to implement flow control at data link layer using SLIDING WINDOW PROTOCOL. Simulate the flow of frames from one node to another.

CODE:

Sender Program:

```
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <unistd.h>

#define MAX 1000

typedef struct {
    int frame_no;
    char data;
} Frame;

int main() {
    char message[MAX];
    int window_size;
    int total_frames;
    int base = 0;
    int ack;
    int i;
```

```
FILE *sender, *receiver;
printf("Enter window size : ");
scanf("%d", &window_size);
getchar();                    //consume newline
printf("Enter Text Message : ");
fgets(message, MAX, stdin);
//Remove newline
message[strcspn(message, "\n")] = '\0';
total_frames = strlen(message);
printf("\n--- Sender started ---\n");
while(base < total_frames) {
    sender = fopen("Sender_Buffer.txt", "w");
    if(sender == NULL) {
        printf("Error opening Sender_Buffer.txt\n");
        return 1;
    }
    printf("\n Sending frames: \n");
    for(i=base; i<base+window_size &&
            i<total_frames; i++) {
        fprintf(sender, "%d %c\n", i,
                message[i]);
        printf("Frame No: %d DATA : %c\n", i,
                message[i]);
    }
```

```
        fclose (sender);

        printf ("\n Waiting for acknowledgement.... \n");
        sleep(3);

    //Read ACK
     receiver = fopen ("Recierer_Buffer.txt", "r");
     if(receiver == NULL){
        printf ("Receiver_Buffer.txt not found. \n");
        return 1;
    }
    if (fscanf (receiver, "%d", &ack) != 1) {
       printf ("Invalid ACK received. \n");
       fclose (receiver);
       continue;
    }
    fclose (receiver);
    printf ("ACK received : %d \n", ack);
    if (ack > base && ack <= total_frames) {
       printf ("ACK is correct. \n");
       base = ack;
       printf ("Moving window forward.... \n");
    }
    else{
       printf ("NACK / Incorrect ACK received. \n");
       printf ("Resending frames from frame %d.... \n", base);
    }
```

```
        sleep(1);
    }
    printf("\n All frames transmitted succesfully.");

// Final Buffer

    sender = fopen("sender_Buffer.txt", "w");
    if (sender != NULL) {
        fprintf(sender, "TRANSMISSION COMPLETE \n");
        fclose(sender);
    }
    return 0;
}
```

Receiver Program:

```
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>

int main() {
    FILE *sender *recierer;
    int frame_no;
    int expected_frame = 0;
    char data;
    printf("--- Receiver Started ---\n");
    while (1) {
        sleep (2);
        sender = fopen("sender_Buffer.txt", "r"
```

```
if (sender == NULL) {
    printf ("Sender_Buffer.txt not found.\n");
    continue;
}
if (fscanf (sender, "%d %c", &frame_no,
                &data) != 2) {
    fclose (sender);
    sleep (1);
    continue;
}
fclose (sender);
// Check frame number
if (frame_no == expected_frame) {
printf ("\n Frame received successfully \n");
printf ("Frame No: %d DATA : %c\n",
                frame_no, data);
expected_frame++;
// Send ACK for the next expected frame
receiver = fopen ("Receiver_Buffer.txt", "w");
if (receiver == NULL) {
    printf ("Error operating Receiver_Buffer.txt", "w
    return 1;
}
fprintf (receiver, "%d\n", expected_frame);
fclose (receiver);
printf ("ACK sent : %d\n", expected_frame);
}
else {
```

```
        printf("\n ERROR : Unexpected Frame \n");
        printf("Expected Frame : %d\n", expected_frame);
        printf("Received Frame : %d\n", frame_no);

        // Send NACK by sending the expected frame no.
        receiver = fopen("Receiver_Buffer.txt", "w");
        if (receiver == NULL) {
            printf("Error opening Receiver_Buffer.txt\n");
            return 1;
        }
        fprintf(receiver, "%d\n", expected_frame);
        fclose(receiver);

        printf("NACK sent for frame : %d\n",
                                    expected_frame);
    }
    if (frame_no < 0)
        break;
}
return 0;
}
```

OUTPUT:

```
--- Sender started ---
Sending Frames :
Frame No: 0     DATA : H
Frame No : 1    DATA : E
Frame No : 2    DATA : L
```

Waiting for acknowledgement...

ACK received : 3

ACK is correct.

Moving window forward...

Sending Frames :

Frame No: 3 DATA: L

Frame No: 4 DATA : O

Waiting for acknowledgement....

ACK received : 5

ACK is correct.

Moving window forward...

All frames transmitted successfully!

RESULT :

Thus, the implementation of flow control at data link layer using SLIDING window protocol is verified successfully.